CS 663: Digital Image Processing

Assignment 1: Problem 2(b) Report

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Contents

1 Thresholding

1.1 2(b) Otsu Thresholding

Implementation

Instead of relying on heuristic guesses, Otsu's method computes the optimal global threshold autonomously. The algorithm exhaustively searches the image's intensity histogram to find the threshold t that minimizes within-class variance, which is mathematically equivalent to maximizing the between-class variance. The objective function maximized by the C++ implementation is:

$$\sigma_b^2 = w_1 w_2 (\mu_1 - \mu_2)^2$$

where w_1 and w_2 are the class probabilities, and μ_1 and μ_2 are the class means for the background and foreground.

Comparative Analysis

Comparing Otsu's computed values against our manually tuned thresholds exposes a fundamental limitation: Otsu operates under the strict mathematical assumption that the image histogram is perfectly bimodal, which fails catastrophically under uneven lighting.

- **Receipt:** Because the histogram is cleanly bimodal, Otsu calculated a highly accurate threshold, resulting in a nearly perfect segmentation that matches the ideal manual tuning.
- **Lilavati Manuscript:** Otsu's variance maximization prioritized separating the dark stains from the bright paper. This mathematically forced the water stains below the threshold, rendering them as massive black obstructions that destroy the surrounding text.
- **Blackboard:** Otsu's computed threshold successfully preserves the faint text on the left but struggles with the illumination gradient, introducing chalk-dust noise artifacts on the brighter right side.
- **QR Code:** Because the "black" pixels in the washed-out top-left corner possess higher absolute intensities than the "white" background pixels in the shadowed bottom-right, Otsu's threshold completely erases the shadowed data. This proves mathematically that no single scalar can segment this image.

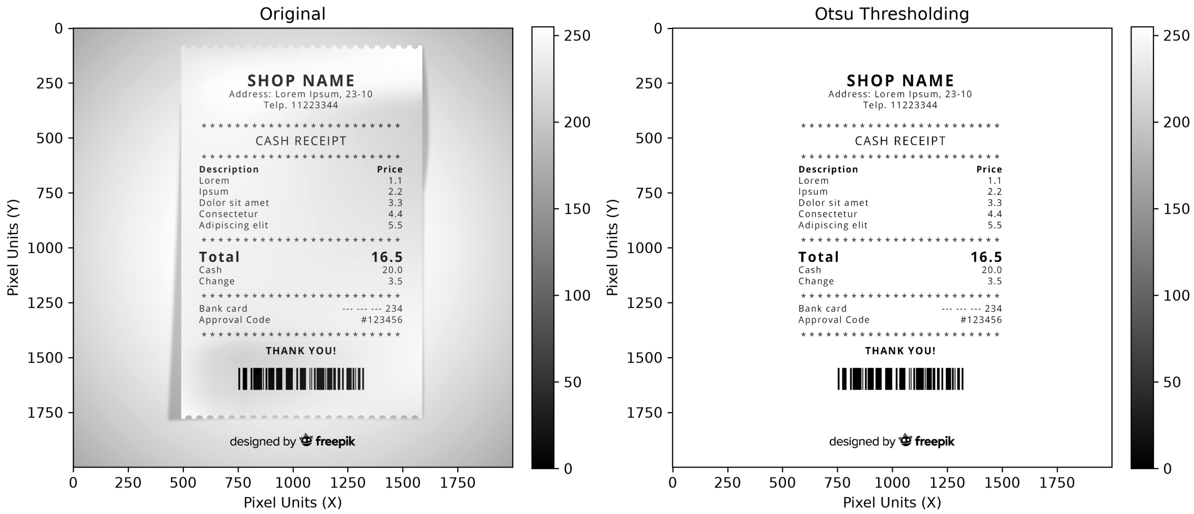

Figure 1: Otsu Thresholding of `receipt.png`.

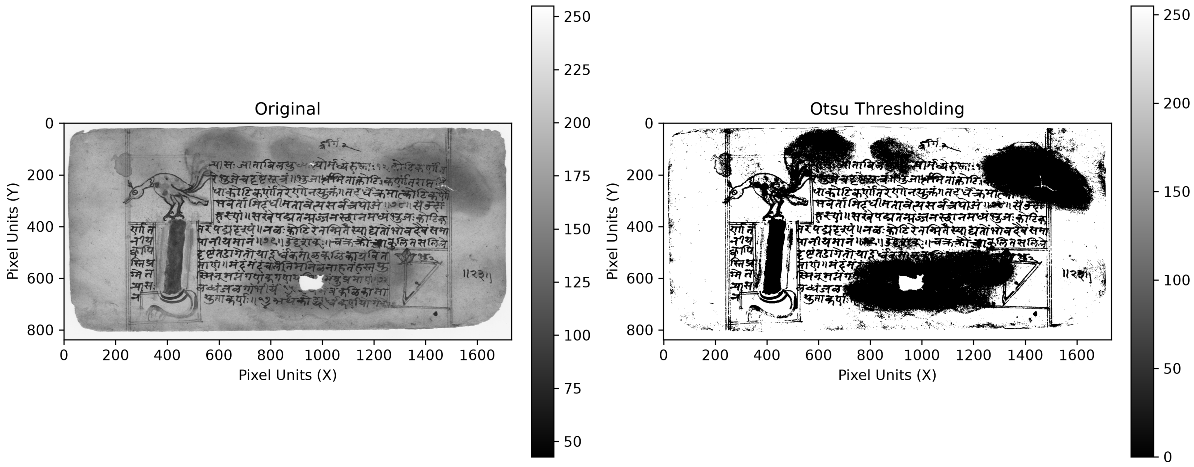

Figure 2: Otsu Thresholding of `lilavati.png`.

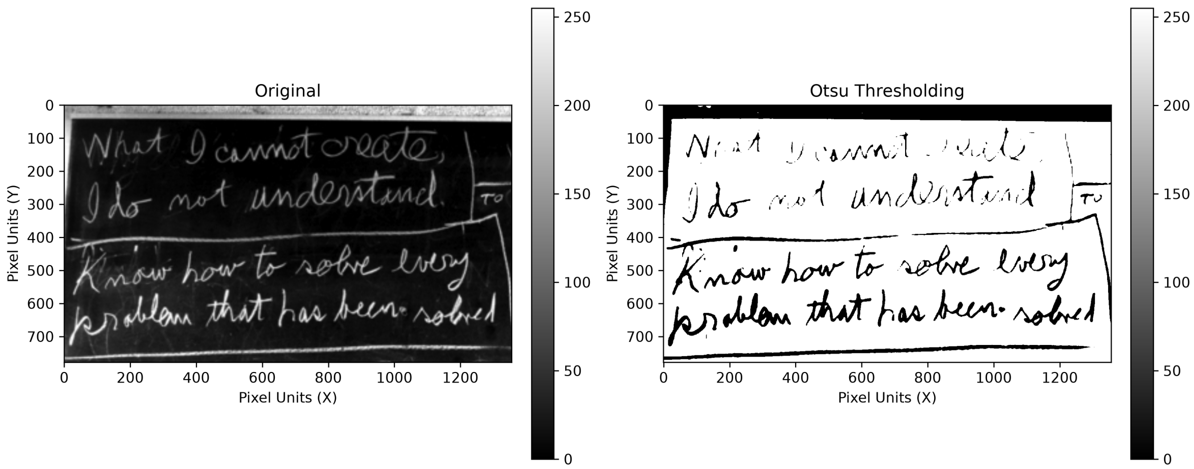

Figure 3: Otsu Thresholding of `blackboard.png` (inverted).

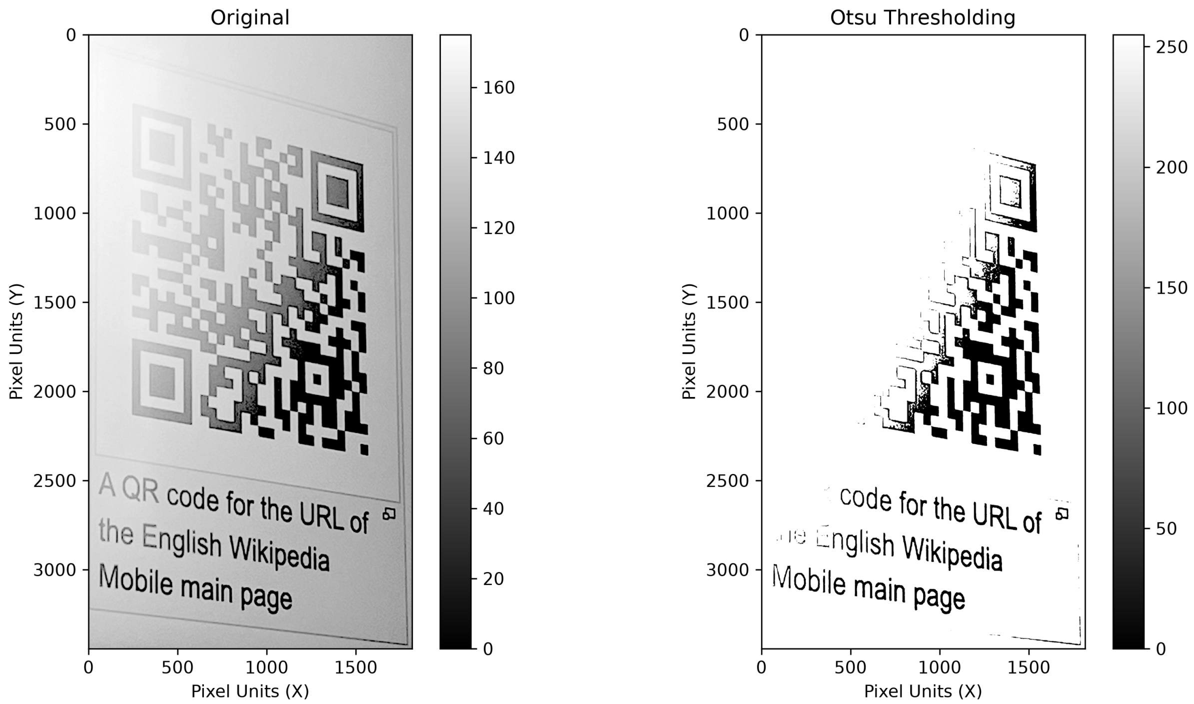

Figure 4: Otsu Thresholding of `qr.png`.